

```

1  #include <stdio.h>
2
3  #define MAX 100
4
5  void main() {
6      int n, i, j;
7
8      int PID[MAX], AT[MAX], ST[MAX], TYPE[MAX];
9      int CT[MAX], TAT[MAX], WT[MAX], Start[MAX];
10
11      int systemQueue[MAX], userQueue[MAX];
12      int sysCount = 0, userCount = 0;
13
14      int time = 0, temp;
15
16      printf("Enter number of processes: ");
17      scanf("%d", &n);
18
19      for(i = 0; i < n; i++) {
20          printf("Process %d\n", i + 1);
21          printf("PID: ");
22          scanf("%d", &PID[i]);
23          printf("Arrival Time: ");
24          scanf("%d", &AT[i]);
25          printf("Burst Time: ");
26          scanf("%d", &ST[i]);
27          printf("Type (0=System, 1=User): ");
28          scanf("%d", &TYPE[i]);
29      }
30
31
32
33      for(i = 0; i < n; i++) {
34          if(TYPE[i] == 0)
35              systemQueue[sysCount++] = i;
36          else
37              userQueue[userCount++] = i;
38      }
39
40
41      for(i = 0; i < sysCount - 1; i++) {
42          for(j = i + 1; j < sysCount; j++) {
43              if(AT[systemQueue[i]] > AT[systemQueue[j]]) {
44                  temp = systemQueue[i];
45                  systemQueue[i] = systemQueue[j];
46                  systemQueue[j] = temp;
47              }
48          }
49      }
50
51      for(i = 0; i < userCount - 1; i++) {
52          for(j = i + 1; j < userCount; j++) {
53              if(AT[userQueue[i]] > AT[userQueue[j]]) {

```

```
multilevel.c - CodeBlocks 25.03
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multilevel.c X
46         systemQueue[j] = temp;
47     }
48 }
49
50
51 for(i = 0; i < userCount - 1; i++) {
52     for(j = i + 1; j < userCount; j++) {
53         if(AT[userQueue[i]] > AT[userQueue[j]]) {
54             temp = userQueue[i];
55             userQueue[i] = userQueue[j];
56             userQueue[j] = temp;
57         }
58     }
59 }
60
61
62 for(l = 0; l < sysCount; l++) {
63     int p = systemQueue[l];
64
65     if(time < AT[p])
66         time = AT[p];
67
68     Start[p] = time;
69     time += BT[p];
70     CT[p] = time;
71
72     TAT[p] = CT[p] - AT[p];
73     WF[p] = TAT[p] - BT[p];
74 }
75
76 for(i = 0; i < userCount; i++) {
77     int p = userQueue[i];
78
79     if(time < AT[p])
80         time = AT[p];
81
82     Start[p] = time;
83     time += BT[p];
84     CT[p] = time;
85
86     TAT[p] = CT[p] - AT[p];
87     WF[p] = TAT[p] - BT[p];
88 }
89
90 printf("\nPID\tAT\tBT\tTYPE\tCT\tTAT\tWF\n");
91 for(i = 0; i < n; i++) {
92     printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
93         PID[i], AT[i], BT[i], TYPE[i],
94         CT[i], TAT[i], WF[i]);
95 }
96
97
98 }
```

C:\Users\BMSCECSE-L3-39\Documents\1vn24c086\week 4\multilevel.c C/C++ Windows (CR-LF) WINDOWS-1252 Line 50, Col 1, Pos 1173 Insert Read/Write default

```
C:\Users\BMSCECESE-L13-39\0 X + - X
Enter number of processes: 5

Process 1
PID: 1
Arrival Time: 0
Burst Time: 4
Type (0=System, 1=User): 0

Process 2
PID: 2
Arrival Time: 1
Burst Time: 2
Type (0=System, 1=User): 1

Process 3
PID: 3
Arrival Time: 2
Burst Time: 3
Type (0=System, 1=User): 1

Process 4
PID: 4
Arrival Time: 2
Burst Time: 2
Type (0=System, 1=User): 0

Process 5
PID: 5
Arrival Time: 8
Burst Time: 5
Type (0=System, 1=User): 0

PID  AT   BT   TYPE  CT   TAT  WT
1     0    4    0      4    4    0
2     1    2    1     15   14   12
3     2    3    1     18   16   13
4     2    2    0      6    4    2
5     8    5    0     13    5    0

Process returned 5 (0x5)  execution time : 106.966 s
Press any key to continue.
```